DSA PROJECT ABSTRACT

Deadline Collision Detector

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1. About the Project

The Deadline Collision Detector is a console program written in C. It helps a student keep track of assignments, lab records, tests and exams, and it warns when too many deadlines fall close together.

For example, a student may have a DLD assignment on Monday, an OOP lab record on Tuesday, a DBMS test on Wednesday, and on Thursday a DLD exam with two assignments. Seeing all this at once is hard. Our program finds such overloaded days and warns the student early, before everything piles up.

We write all the data structures ourselves in C, instead of using libraries that hide how they work. This way we learn the DSA properly.

2. Features

- **Add and manage deadlines:** The user enters the task name, subject, due date and task type. Tasks can also be viewed, edited or deleted.
- **Collision alerts:** The program warns when many tasks fall on the same day, or when three or more tasks fall within 24 hours.
- **Urgency ranking:** The nearest deadlines are shown first, and overdue tasks are highlighted.
- **Permanent storage:** Tasks are saved in a text file, so they are still there after closing the program.

3. Data Structures Used

- **Linked list** stores and manages the task records.
- **Binary search tree (BST)** keeps tasks arranged by due date, for searching and displaying deadlines.
- **Min-heap** gives the nearest upcoming deadline quickly.
- **Sorting** ranks tasks by urgency or remaining time.
- **File handling** saves tasks between runs of the program.

4. Main Menu

```
===== DEADLINE COLLISION DETECTOR =====

1. Add a deadline
2. View all deadlines
3. Search for a deadline
4. Edit a deadline
5. Delete a deadline
6. Check deadline collisions
7. View nearest upcoming deadlines
8. Sort tasks by urgency
9. Save and exit

Enter your choice:
```

5. Project Scope

- It is a console-based C application. A better interface can be added later only if needed.
- Dates are entered in DD/MM/YYYY format, for example 25/10/2026.
- Collision check: more than one task on the same day, and three or more tasks within 24 hours.
- Tasks are saved in a simple text file.
- We will build step by step: first a simple version, then add each data structure one by one.

6. Tools Used

C language, GCC compiler, and VS Code or Code::Blocks.

7. Expected Outcome

A working program that helps students plan their work and avoid last-minute pressure, and shows how linked lists, BST, heaps and sorting are used in a real problem.